

PDF Implementation Process

Libraries Downloaded:

- jsPDF: The core library used for generating PDF documents from JavaScript.
- jsPDF-autotable: A plugin for jsPDF that handles drawing and formatting data tables.

The Step-by-Step Process:

1. Library Installation: I used the Node Package Manager (npm) to download and install both 'jsPDF' and 'jsPDF-autotable' into the project.
2. PDF Document Setup: I created a utility function that initializes a new PDF document. Because there are 28 fields, I configured the PDF to be in landscape orientation and used an A3 page size to ensure all the columns would fit without overlapping.
3. Data Structuring: I mapped over the employee data to extract the 28 required fields. For fields that can have multiple items (like RAM sticks, Storage drives, and GPUs), I programmed it to combine all the items using line breaks so they display neatly inside a single cell for that specific employee.
4. Table Generation: I fed the structured headers and the mapped employee data into the 'autotable' plugin. I customized the table theme by coloring the headers dark blue (to match the web app's theme), reducing the font size, and enabling automatic line breaking for long text.
5. Interface Integration: Finally, I updated the main application interface by adding an 'Export PDF' button to the toolbar. This button takes whatever list of devices is currently filtered on your screen and triggers the utility function to download the customized PDF file to your computer.